Unit 1 Progress Check: FRQ Part C

This question involves processing an email address to create three usernames.

The email address is stored in a String variable named email, which consists of a first name, followed by ".", followed by a last name, followed by "@", followed by a domain name. Assume that the email address is stored in this format and the last name has at least three characters.

The usernames will be formatted as follows.

- Username 1 will consist of the first name, followed by "_", followed by the last name.
- Username 2 will consist of the first initial of the first name, followed by "_", followed by the last name.
- Username 3 will consist of a randomly-selected character from the first name , followed by "_", followed by a sequence of three consecutive characters selected from a random position in the last name.

Example 1

Suppose email is equal to "joseph.martin@school.org".

- Username 1 will be "joseph_martin".
- Username 2 will be "j_martin".
- Possible values of Username 3 include "s_art" and "p_tin".

Example 2

Suppose email is equal to "virginia.farmingham@school.net".

- Username 1 will be "virginia_farmingham".
- Username 2 will be "v_farmingham".
- Possible values of Username 3 include "g_ham" and "i_arm".

1. Write a statement that uses the String method indexOf to find the index of the first "." in email and assign it to a variable. Write another statement that uses the String method indexOf to find the index of "@" and assign it to a variable.

Part A Point 1

Select a point value to view scoring criteria or to score the response.

To earn this point, responses must demonstrate the following:

Unit 1 Progress Check: FRQ Part C

- Finds the positions of the first period `"."` and the `"@"` using appropriate `String` method calls

Canonical Solution:

```
int periodLoc = email.indexOf(".");
int atLoc = email.indexOf("@");
```

0	1 ✓

The response earns the point.

Responses will not earn the point if they

- make any incorrect call to a `String` method

Unit 1 Progress Check: FRQ Part C

2. Write a statement that uses the String method substring to extract the first name from email and assign it to a variable. Write another statement that uses the String method substring to extract the last name from email and assign it to a variable. Then use the results of those statements to create and assign the correct value to a String variable for username 1.

Part B Point 1

Select a point value to view scoring criteria or to score the response.

To earn this point, responses must demonstrate the following:

- Extracts the first and last names from the email address

Canonical Solution:

```
String first = email.substring(0, periodLoc);
String last = email.substring(periodLoc + 1, atLoc);
String user1 = first + "_" + last;
```

0	1 ✓

The response earns the point.

Responses can still earn the point even if they

- incorrectly identify the location of the first period `"."` or the `"@"`

Responses will not earn the point if they

- make any incorrect call to a `String` method

Part B Point 2

Select a point value to view scoring criteria or to score the response.

To earn this point, responses must demonstrate the following:

- Creates a string variable that contains first name and last name

Canonical Solution:

```
String first = email.substring(0, periodLoc);
String last = email.substring(periodLoc + 1, atLoc);
String user1 = first + "_" + last;
```

Unit 1 Progress Check: FRQ Part C

0	1 ✓

The response earns the point.

Responses can still earn the point even if they

- omit the underscore (`"_"`)

Responses will not earn the point if they

- make any incorrect call to a `String` method

Unit 1 Progress Check: FRQ Part C

3. Write a statement that uses the String method substring to extract the first initial of the first name from email and assign it to a variable. Use the results of this statement and the statements you wrote in part B to create and assign the correct value to a String variable for username 2.

Part C Point 1

Select a point value to view scoring criteria or to score the response.

To earn this point, responses must demonstrate the following:

- Extracts the first initial of the first name

Canonical Solution:

```
String firstInit = email.substring(0, 1);
String user2 = firstInit + "_" + last;
```

0	1 ✓

The response earns the point.

Responses will not earn the point if they

- make any incorrect call to a `String` method

Part C Point 2

Select a point value to view scoring criteria or to score the response.

To earn this point, responses must demonstrate the following:

- Creates a string variable that contains the first initial of the first name and the last name

Canonical Solution:

```
String firstInit = email.substring(0, 1);
String user2 = firstInit + "_" + last;
```

0	1 ✓

The response earns the point.

Responses can still earn the point even if they

Unit 1 Progress Check: FRQ Part C

- omit the underscore (`"_"`)

Responses will not earn the point if they

- make any incorrect call to a `String` method

Unit 1 Progress Check: FRQ Part C

4. Write a statement that uses the Math method random to generate a random number between 0 and the length of the first name. Write another statement that uses the Math method random to generate a random number between 0 and the length of the last name minus 2.

Part D Point 1

Select a point value to view scoring criteria or to score the response.

To earn this point, responses must demonstrate the following:

- Uses the `random` method to generate a random number between 0 and the length of the first name

Canonical Solution:

```
int lenFirst = first.length();
int numFirst = (int)(Math.random() * lenFirst);
int lenLast = last.length() - 2;
int numLast = (int)(Math.random() * lenLast);
```

0	1 ✓

The response earns the point.

Responses will not earn the point if they

- make any incorrect call to the `random` method

Part D Point 2

Select a point value to view scoring criteria or to score the response.

To earn this point, responses must demonstrate the following:

- Uses the `random` method to generate a random number between 0 and the length of the last name minus 2

Canonical Solution:

```
int lenFirst = first.length();
int numFirst = (int)(Math.random() * lenFirst);
int lenLast = last.length() - 2;
int numLast = (int)(Math.random() * lenLast);
```

0	1 ✓

Unit 1 Progress Check: FRQ Part C

The response earns the point.

Responses will not earn the point if they

- make any incorrect call to the `random` method

Unit 1 Progress Check: FRQ Part C

5. Write a code segment that uses the random numbers that you generated in part D to create and assign the correct value to a String variable for username 3.

Part E Point 1

Select a point value to view scoring criteria or to score the response.

To earn this point, responses must demonstrate the following:

- Extracts a randomly selected character from the first name and randomly selected sequence of 3 characters from the last name

Canonical Solution:

```
String firstChar = first.substring(numFirst, numFirst + 1);
String lastChar3 = last.substring(numLast, numLast + 3);
String user3 = firstChar + "_" + lastChar3;
```

0	1 ✓

The response earns the point.

Responses can still earn the point even if they

- incorrectly generate a random number

Responses will not earn the point if they

- make any incorrect call to a `String` method

Part E Point 2

Select a point value to view scoring criteria or to score the response.

To earn this point, responses must demonstrate the following:

- Creates a string variable containing identified character from the first name and identified characters from the last name

Canonical Solution:

```
String firstChar = first.substring(numFirst, numFirst + 1);
String lastChar3 = last.substring(numLast, numLast + 3);
String user3 = firstChar + "_" + lastChar3;
```

Unit 1 Progress Check: FRQ Part C

0	1 ✓

The response earns the point.

Responses can still earn the point even if they

- omit the underscore (`"_"`)
- incorrectly obtain the character from the first name or the three characters from the last name

Responses will not earn the point if they

- make any incorrect call to a `String` method